Sunways PV Inverter
Modbus RTU Protocol

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Version 00.07

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Change History

Version	Date	Signature	Notice
00.01	20190918	Lee	Initial release.
00.02	20191013	Lee	Added some register data description; Fixed some data type errors; Deleted invalid register address; Added inverter start-stop setting register
00.03	20191025	Lee	Finished the Form3.2, Model Id;
00.04	20191029	Lee	Added P_AC
00.05	20200731	Lee	Updated model information
00.06	20211019	Neo	Table 3.1 add register information; Table 3.2 add model information
00.07	20230706	Hu	Table 4.1 add model information

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1 Introduction

Modbus-RTU protocol is a widely used communication protocol in industry and is a common language used in electrical communication terminals. Through this protocol, inverters can communicate with each other via network (such as RS485 bus) and other devices. It has become a universal industry standard. With it, inverter equipment manufactured by different manufacturers can be connected into an industrial network for centralized monitoring. The protocol describes how the master and slave nodes are defined, how the master node accesses other devices by using various request methods, how the slave nodes respond to requests from other devices, and how both parties detect and record errors. It defines the message domain pattern and the data content in detail.

In this paper, the Modbus protocol of the Sunways inverter is described and explained, which is used to standardize and constrain the subsequent third-party integration development and customization.

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2 Communication Protocol Summary

Modbus communication protocol includes physical layer, data link layer and application layer.

2.1 Physical Layer

- Communicate with two - line RS-485 via serial port.
- The baud rate is 9600.
- Adopt RTU mode transmission, asynchronous mode.
- One starting bit.
- Eight data bits.
- No verification.
- 1 stop bit.

2.2 DLL

2.2.1 Addressing Mode

The protocol supports unicast and broadcast, and the address assignment rules are as follows:

Broadcast Address	Slave Node Address	Retain
0	1~247	248~255

2.2.2 Frame Structure

Address	Function Code	Data	CRC Check Code
1 byte	1 byte	2*N bytes	2 bytes

Note:

- The maximum frame length is no more than 256 bytes.
- CRC check code low byte before, high byte after.
- The definition of all frame structures in this paper contains only the function codes and data sections.

2.2.3 Data Coding

Modbus uses a 'big- endian' to represent addresses and data items. This means that when sending more than one byte, the most significant bit is sent first, such as:

Register size	Value

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16 bits	0xABCD

The first byte sent is 0xAB and the second byte is 0xCD.

2.2.4 Interactive Process

In any case, the communication process is initiated by the master node and the slave node does not initiate the communication actively.

In unicast mode, the slave node responds to the master node's command in a question-and-answer manner. If the master node does not receive a reply from the slave node within 5 s, the communication timeout is considered.

In broadcast mode, the slave node only receives commands issued by the master node and does not respond to command frames issued by the master node.

2.2.5 CRC Verify

```
const INT8U auchCRCHi[256] = {    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0,
0x80, 0x41, 0x00, 0xC1, 0x81, 0x40,
                    0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41,
                    0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41,
                    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40,
                    0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41,
                    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40,
                    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40,
                    0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41,
                    0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41,
                    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40,
                    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40,
                    0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41,
                    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40,
                    0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41,
```

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```
                    0x01, 0xC0, 0x80, 0x41, 0x00, 0xC1, 0x81, 0x40, 0x00, 0xC1, 0x81, 0x40,
0x01, 0xC0, 0x80, 0x41,
                    0x00, 0xC1, 0x81, 0x40, 0x01, 0xC0, 0x80, 0x41, 0x01, 0xC0, 0x80, 0x41,
0x00, 0xC1, 0x81, 0x40
                    };

const INT8U auchCRCLo[256]    = { 0x00, 0xC0, 0xC1, 0x01, 0xC3, 0x03, 0x02, 0xC2, 0xC6,
0x06, 0x07, 0xC7, 0x05, 0xC5, 0xC4, 0x04,
                    0xCC, 0x0C, 0x0D, 0xCD, 0x0F, 0xCF, 0xCE, 0x0E, 0x0A, 0xCA, 0xCB, 0x0B,
0xC9, 0x09, 0x08, 0xC8,
                    0xD8, 0x18, 0x19, 0xD9, 0x1B, 0xDB, 0xDA, 0x1A, 0x1E, 0xDE, 0xDF, 0x1F,
0xDD, 0x1D, 0x1C, 0xDC,
                    0x14, 0xD4, 0xD5, 0x15, 0xD7, 0x17, 0x16, 0xD6, 0xD2, 0x12, 0x13, 0xD3,
0x11, 0xD1, 0xD0, 0x10,
                    0xF0, 0x30, 0x31, 0xF1, 0x33, 0xF3, 0xF2, 0x32, 0x36, 0xF6, 0xF7, 0x37,
0xF5, 0x35, 0x34, 0xF4,
                    0x3C, 0xFC, 0xFD, 0x3D, 0xFF,    0x3F, 0x3E, 0xFE, 0xFA, 0x3A, 0x3B,
0xFB, 0x39, 0xF9, 0xF8, 0x38,
                    0x28, 0xE8, 0xE9, 0x29, 0xEB, 0x2B, 0x2A, 0xEA, 0xEE, 0x2E, 0x2F, 0xEF,
0x2D, 0xED, 0xEC, 0x2C,
                    0xE4, 0x24, 0x25, 0xE5, 0x27, 0xE7, 0xE6, 0x26, 0x22, 0xE2, 0xE3, 0x23,
0xE1, 0x21, 0x20, 0xE0,
                    0xA0, 0x60, 0x61, 0xA1, 0x63, 0xA3, 0xA2, 0x62, 0x66, 0xA6, 0xA7, 0x67,
0xA5, 0x65, 0x64, 0xA4,
                    0x6C, 0xAC, 0xAD, 0x6D, 0xAF, 0x6F, 0x6E, 0xAE, 0xAA, 0x6A, 0x6B,
0xAB, 0x69, 0xA9, 0xA8, 0x68,
                    0x78, 0xB8, 0xB9, 0x79, 0xBB, 0x7B, 0x7A, 0xBA, 0xBE, 0x7E, 0x7F, 0xBF,
0x7D, 0xBD, 0xBC, 0x7C,
                    0xB4, 0x74, 0x75, 0xB5, 0x77, 0xB7, 0xB6, 0x76, 0x72, 0xB2, 0xB3, 0x73,
0xB1, 0x71, 0x70, 0xB0,
                    0x50, 0x90, 0x91, 0x51, 0x93, 0x53, 0x52, 0x92, 0x96, 0x56, 0x57, 0x97, 0x55,
0x95, 0x94, 0x54,
                    0x9C, 0x5C, 0x5D, 0x9D, 0x5F, 0x9F, 0x9E, 0x5E, 0x5A, 0x9A, 0x9B, 0x5B,
0x99, 0x59, 0x58, 0x98,
                    0x88, 0x48, 0x49, 0x89, 0x4B, 0x8B, 0x8A, 0x4A, 0x4E, 0x8E, 0x8F, 0x4F,
0x8D, 0x4D, 0x4C, 0x8C,
                    0x44, 0x84, 0x85, 0x45, 0x87, 0x47, 0x46, 0x86, 0x82, 0x42, 0x43, 0x83, 0x41,
0x81, 0x80, 0x40
                    };

INT16U CRC16(INT8U *puchMsg, INT16U usDataLen)
{
    INT8U uchCRCHi = 0xFF ;
    INT8U uchCRCLo = 0xFF ;
```

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```
    INT8U uIndex=0 ;
    while (usDataLen--)
    {
        uIndex = uchCRCLo ^ *puchMsg++;
        uchCRCLo = uchCRCHi ^ auchCRCHi[uIndex];
        uchCRCHi = auchCRCLo[uIndex];
    }
    return (((INT16U)uchCRCHi << 8) | (INT16U)uchCRCLo) ;
}
```

2.3 Application Layer

MODBUS has three types of function codes.

Common function codes: They are well defined function codes that are guaranteed to be unique, modifiable by MODBUS organization, publicly proven, with available conformance tests, proven in MB IETF RFC, and contain defined common assigned function codes and unassigned reserved function codes for future use.

User-defined function codes: There are two defined scopes of user-defined function codes, namely 65 to 72 and 100 to 110. Users can select and implement a function code without any approval from MODBUS organization, and the use of the selected function code cannot be guaranteed to be unique.

Preserve function codes: Function codes that are commonly used by some companies for traditional products and that are not valid for public uses.

The common function codes used in this Agreement are as follows:

Function Code	Function Description
03（0x03）	Read hold register values
06（0x06）	Write Single Register
16（0x10）	Write Multiple Registers

Abbreviations and terms are defined as follows:

Designation	Description
Register address	One address corresponds to one 2Bytes data
U16	Unsigned 16-bit reshaping data
U32	Unsigned 32-bit reshaping data
I16	Signed 16-bit reshaping data
I32	Signed 32-bit reshaping data
STR	Character string
N/A	None

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2.3.1 Function Code03(0x03)

The master node requests the frame format：

Data domain	Length(byte)	Description
Slave node address	1	1~247
Function code	1	0x03
Register start address	2	0x0000~0xFFFF
Register number	2	1~125
CRC	2	N/A

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The slave node responds normally to the frame format

Data domain	Length(byte)	Description
Slave node address	1	1~247
Function code	1	0x03
Number of bytes	1	2*N
Register values	2*N	N/A
CRC	2	N/A

The slave node exception responds to the frame format

Data domain	Length(byte)	Description
Slave node address	1	1~247
Function code	1	0x83
exception code	1	1/2/3
CRC	2	N/A

2.3.2 Function Code06(0x06)

The master node requests the frame format：

Data domain	Length(byte)	Description
Slave node address	1	1~247
Function code	1	0x06
Register address	2	0x0000~0xFFFF
Register value	2	0x0000~0xFFFF
CRC	2	N/A

The slave node responds normally to the frame format

Data domain	length(byte)	Description
Slave node address	1	1~247
Function code	1	0x06
Register address	2	0x0000~0xFFFF
Register value	2	0x0000~0xFFFF
CRC	2	N/A

The slave node exception responds to the frame format

Data domain	length(byte)	Description
Slave node address	1	1~247
Function code	1	0x86
Exception code	1	1/2/3
CRC	2	N/A

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2.3.3 Function Code16(0x10)

The master node requests the frame format:

Data Domain	Length(byte)	Description
Slave node address	1	0~247
Function code	1	0x10
Register address	2	0x0000~0xFFFF
Number of registers	2	0x0000~0x007b
Number of bytes	1	2*N
Register values	2*N	Value
CRC	2	N/A

"N" is the number of registers(0x0000~0x007b)

The slave node responds normally to the frame format

Data Domain	Length(byte)	Description
Slave node address	1	1~247
Function code	1	0x10
Register address	2	0x0000~0xFFFF
Number of registers	2	0x0000~0x007b
CRC	2	N/A

The slave node exception responds to the frame format

Data Domain	Length(byte)	Description
Slave node address	1	1~247
Function code	1	0x90
Exception code	1	1/2/3
CRC	2	N/A

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3 Register Table

Table 3.1

SN	Register Address	Words	Data Description	Read-write	Types	Units	Gain	Notes
1	10000	8	Serial number of devices	RO	STR	N/A	1	Resolves to characters in high and low bytes
	.....							
2	10008	1	Model information	RO	U16	N/A	1	See table 3.2 for information resolution
3	10009	1	Output mode	RO	U16	N/A	1	0: three phase four-wire 1 : three-phase three-wire
4	10010	1	Communication protocol version	RO	U16	N/A	1	Current version 0
5	10011	2	Firmware version number	RO	U32	N/A	1	Byte parsing
	10012							
6	10100	1	Time: year/month	RO	U16	N/A	1	The data is parsed in terms of high and low bytes
7	10101	1	Time: day/hour	RO	U16	N/A	1	
8	10102	1	Time: minute/second	RO	U16	N/A	1	
9	10104	1	Safety standard / Area code	RO	U16	N/A	1	
10	10105	1	Working state of inverter	RO	U16	N/A	1	0: wait, Waiting for the grid 1: check, self-inspection 2: normal, generating 3: fault, device failure 4: flash, firmware update
11	10112	2	Fault FLAG1	RO	U32	N/A	1	The bitwise resolution is shown in Table 3.3
	10113							
12	10114	2	Fault FLAG2	RO	U32	N/A	1	
	10115							
13	10120	2	Fault FLAG3	RO	U32	N/A	1	
	10121							
14	11006	1	Grid LINE AB voltage	RO	U16	V	10	
15	11007	1	Grid LINE BC voltage	RO	U16	V	10	
16	11008	1	Grid LINE CA voltage	RO	U16	V	10	
17	11009	1	Grid A phase voltage	RO	U16	V	10	

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Continued Table 3.1

SN	Register Address	Words	Data Description	Read-write	Types	Units	Gain	Notes
18	11010	1	Grid A phase current	RO	U16	A	10	
19	11011	1	Grid B phase voltage	RO	U16	V	10	
20	11012	1	Grid B phase current	RO	U16	A	10	
21	11013	1	Grid C phase voltage	RO	U16	V	10	
22	11014	1	Grid C phase current	RO	U16	A	10	
23	11015	1	Grid frequency	RO	U16	Hz	100	
24	11016	2	P_AC	RO	U32	kW	1000	
	11017							
25	11018	2	Daily generating capacity	RO	U32	kWh	10	
	11019							
26	11020	2	Total generation	RO	U32	kWh	10	
	11021							
27	11022	2	Total generating time	RO	U32	H	1	
	11023							
28	11024	2	Output apparent power	RO	U32	kVar	1000	
	11025							
29	11026	2	Output reactive power	RO	I32	kVar	1000	
	11027							
30	11028	2	Total power input	RO	U32	kW	1000	
	11029							
31	11030	1	Power factor	RO	I16	N/A	1000	
32	11031	1	Efficiency	RO	U16	%	100	
33	11032	1	Temperature 1	RO	I16	℃	10	
34	11033	1	Temperature 2	RO	I16	℃	10	
35	11034	1	Temperature 3	RO	I16	℃	10	
36	11035	1	Temperature 4	RO	I16	℃	10	
37	11036	1	BUS Voltage	RO	U16	V	10	
38	11037	1	NBS Voltage	RO	U16	V	10	
39	11038	1	PV1 voltage	RO	U16	V	10	
40	11039	1	PV1 current	RO	U16	A	10	
41	11040	1	PV2 voltage	RO	U16	V	10	

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Continued Table 3.1

SN	Register Address	Words	Data Description	Read-write	Types	Units	Gain	Notes
42	11041	1	PV2 current	RO	U16	A	10	
43	11042	1	PV3 voltage	RO	U16	V	10	
44	11043	1	PV3 current	RO	U16	A	10	
45	11044	1	PV4 voltage	RO	U16	V	10	
46	11045	1	PV4 current	RO	U16	A	10	
47	11046	1	PV5 voltage	RO	U16	V	10	
48	11047	1	PV5 current	RO	U16	A	10	
49	11048	1	PV6 voltage	RO	U16	V	10	
50	11049	1	PV6 current	RO	U16	A	10	
51	11050	1	String 1 current	RO	U16	A	10	
52	11051	1	String 2 current	RO	U16	A	10	
53	11052	1	String 3 current	RO	U16	A	10	
54	11053	1	String 4 current	RO	U16	A	10	
55	11054	1	String 5 current	RO	U16	A	10	
56	11055	1	String 6 current	RO	U16	A	10	
57	11056	1	String 7 current	RO	U16	A	10	
58	11057	1	String 8 current	RO	U16	A	10	
59	11058	1	String 9 current	RO	U16	A	10	
60	11059	1	String 10 current	RO	U16	A	10	
61	11060	1	String 11 current	RO	U16	A	10	
62	11061	1	String 12 current	RO	U16	A	10	
63	11062	2	PV1 input power	RO	U32	kW	1000	
	11063							
64	11064	2	PV2 input power	RO	U32	kW	1000	
	11065							
65	11066	2	PV3 input power	RO	U32	kW	1000	
	11067							
66	11068	2	PV4 input power	RO	U32	kW	1000	
	11069							
67	11070	2	PV5 input power	RO	U32	kW	1000	
	11071							
68	11072	2	PV6 input power	RO	U32	kW	1000	
	11073							
69	11074	1	PV7 voltage	RO	U16	V	10	
70	11075	1	PV7 current	RO	U16	A	10	
71	11076	1	PV8 voltage	RO	U16	V	10	
72	11077	1	PV8 current	RO	U16	A	10	
73	11078	1	PV9 voltage	RO	U16	V	10	
74	11079	1	PV9 current	RO	U16	A	10	
75	11080	1	PV10 voltage	RO	U16	V	10	

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Continued Table 3.1

SN	Register Address	Words	Data Description	Read-write	Types	Units	Gain	Notes
76	11081	1	PV10 current	RO	U16	V	10	
77	11082	1	String 13 current	RO	U16	A	10	
78	11083	1	String 14 current	RO	U16	A	10	
79	11084	1	String 15 current	RO	U16	A	10	
80	11085	1	String 16 current	RO	U16	A	10	
81	11086	1	String 17 current	RO	U16	A	10	
82	11087	1	String 18 current	RO	U16	A	10	
83	11087	1	String 18 current	RO	U16	A	10	
84	11088	1	String 19 current	RO	U16	A	10	
85	11089	1	String 20 current	RO	U16	A	10	
86	11090	2	PV7 input power	RO	U32	kW	1000	
87	11091							
88	11092	2	PV8 input power	RO	U32	kW	1000	
89	11093							
90	11094	2	PV9 input power	RO	U32	kW	1000	
91	11095							
92	11096	2	PV10 input power	RO	U32	kW	1000	
	11097							
93	20000	1	Device real-time clock time	RW	U16	N/A	1	Year Month
	20001	1		RW	U16	N/A	1	Day Hour
	20002	1		RW	U16	N/A	1	Minute Second
94	25008	1	Switch on/off settings	RW	U16	N/A	1	BIT0: Set 1， turn on; Set 0， turn off; BIT1: Set 1， restart;
95	25112	2	Active power limit	WO	U32	W	1	
	25113							
96	25114	1	Active power limit percentage	WO	U16	N/A	1000	[0,1000]
97	25116	2	Reactive power limit	WO	I32	W	1	
	25117							
98	25118	1	Reactive power limit percentage	WO	I16	N/A	1000	[-600,+600]
99	25120	1	PF Settings	RW	I16	N/A	1000	(- 1000,-800]U[800,1000]

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Table 3.2

Model information (10008 high type)		**Single-phase Dual-MPPT PV inverter**	**Single-phase Single-MPPT PV inverter**	**Single-phase Dual-MPPT PV inverter**	**Three-phase Dual-MPPT PV inverter**
		01	02	03	12
Rating information (10008 low byte)	00	STS-3KTL(-P)	N/A	STS-7KTL	STT-6KTL(-P)
	01	STS-3.6KTL(-P)	STS-1KTL-S	STS-8KTL	STT-8KTL(-P)
	02	STS-4.2KTL(-P)	STS-1.5KTL-S	STS-9KTL	STT-10KTL(-P)
	03	STS-4.6KTL(-P)	STS-2KTL-S	STS-10KTL	STT-12KTL(-P)
	04	STS-5KTL(-P)	STS-2.5KTL-S	STS-11KTL	STT-15KTL(-P)
	05	STS-6KTL(-P)	STS-3KTL-S	N/A	STT-17KTL(-P)
	06	N/A	STS-3.3KTL-S	N/A	STT-20KTL(-P)
	07	N/A	STS-1KTL-S-P	N/A	STT-25KTL(-P)
	08	N/A	STS-1.5KTL-S-P	N/A	STT-4KTL(-P)
	09	N/A	STS-2KTL-S-P	N/A	STT-5KTL(-P)
	10		STS-2.5KTL-S-P		
	11		STS-3KTL-S-P		
	12		STS-3.3KTL-S-P		
Model information (10008 high type)		**Three-phase Four-MPPT PV inverter**	**Three-phase Six-MPPT PV inverter**	**Three-phase Ten-MPPT PV inverter**	**N/A**
		16	20	21	N/A
Rating information (10008 low byte)	0	STT-29.9KTL	STT-50KTL(-P)	STT-80KTL	N/A
	1	STT-30KTL	STT-60KTL(-P)	STT-90KTL	N/A
	2	STT-33KTL	N/A	STT-100KTL	N/A
	3	STT-36KTL	N/A	STT-110KTL	N/A
	4	STT-40KTL	N/A	N/A	N/A
	5	STT-45KTL	N/A	N/A	N/A
	6	STT-50KTL-M	N/A	STT-100KTL-HV	N/A
	7	STT-60KTL-M	N/A	STT-110KTL-HV	N/A
	8	N/A	N/A	STT-125KTL-HV	N/A
	9	N/A	N/A		N/A
	10	N/A	N/A		N/A
	11	STT-15KTL-LV			N/A
	12	STT-20KTL-LV			N/A
	13	STT-25KTL-LV			N/A

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Table 3.3

Fault Code	Register Address	Bit	Hexadecimal	Decimal	The Fault Name	中文显示	English Display
1	10112 (Fault FLAG1)	BIT0	0x00000001	1	Grid loss	电网丢失	Mains Lost
2		BIT1	0x00000002	2	Grid voltage Fault	电网电压异常	Grid Voltage Fault
3		BIT2	0x00000004	4	Grid Frequency Fault	电网频率异常	Grid Frequency Fault
4		BIT3	0x00000008	8	DCI Fault	直流分量超限	DCI Fault
5		BIT4	0x00000010	16	ISO Over Limitation	绝缘阻抗超限	ISO Over Limitation
6		BIT5	0x00000020	32	GFCI Fault	漏电流超限	GFCI Fault
7		BIT6	0x00000040	64	PV Over Voltage	输入电压超限	PV Over Voltage
8		BIT7	0x00000080	128	Bus Voltage Fault	母线电压超限	Bus Voltage Fault
9		BIT8	0x00000100	256	Inverter Over Temperature	设备温度超限	Inverter Over Temperature
34	10112 (Fault FLAG2)	BIT1	0x00000002	2	SPI Fault	内部通讯故障二	SPI Fault
35		BIT2	0x00000004	4	E2 Fault	存储器一故障	E2 Fault
36		BIT3	0x00000008	8	GFCI Device Fault	漏电流传感器故障	GFCI Device Fault
37		BIT4	0x00000010	16	AC Transducer Fault	输出电流传感器故障	AC Transducer Fault
38		BIT5	0x00000020	32	Relay Check Fail	继电器故障	Relay Check Fail
39		BIT6	0x00000040	64	Internal Fan Fault	内部风扇故障	Internal Fan Fault
40		BIT7	0x00000080	128	External Fan Fault	外部风扇故障	External Fan Fault
65	10120 (Fault FLAG3)	BIT0	0x00000001	1	SCI Fault	内部通讯故障一	SCI Fault
66		BIT1	0x00000002	2	FLASH Fault	存储器二故障	FLASH Fault

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